

Suppression of wheat blast resistance by an effector of *Pyricularia oryzae* is counteracted by a host specificity resistance gene in wheat

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Summary

- Wheat blast caused by the *Triticum* pathotype of *Pyricularia oryzae* poses a serious threat to wheat production in South America and Asia and is now becoming a pandemic disease. Here, we show that *Rmg8*, a promising wheat gene for resistance breeding, is suppressed by *PWT4*, an effector gene of *P. oryzae*, and in turn that the suppression is counteracted by *Rwt4*, a wheat gene recognizing *PWT4*.
- When *PWT4* was introduced into a wheat blast isolate carrying *AVR-Rmg8* (an avirulence gene corresponding to *Rmg8*), *PWT4* suppressed wheat resistance conferred by *Rmg8*. *PWT4* did not alter the expression of *AVR-Rmg8*, but higher expression of *PWT4* led to more efficient suppression. This suppression was observed in *rwt4* carriers, but not in *Rwt4* carriers, indicating that it is counteracted by *Rwt4*.
- *PWT4* was assumed to have been horizontally transferred from a weed-associated cryptic species, *P. pennisetigena*, to an *Avena* isolate of *P. oryzae* in Brazil. This implies a potential risk of the acquisition of *PWT4* by the wheat blast fungus and the 'breakdown' of *Rmg8*.
- We suggest that *Rmg8* should be introduced together with *Rwt4* into a wheat cultivar when it is used for resistance breeding.

Introduction

Plant diseases pose significant challenges to food security as exemplified by outbreaks of wheat stem rust 'Ug99' race groups in Africa and the Middle East (Singh *et al.*, 2015) and Asian soybean rust in Brazil (Godoy *et al.*, 2016). Resistance breeding is an environmentally friendly measure to control plant diseases. Specificity between pathogen variants (races) and plant genotypes (cultivars) follows gene-for-gene interactions in which an outcome of interactions is conditioned by alleles of a gene conditioning resistance (*R* gene) in the plant and alleles of its corresponding gene conditioning avirulence (*AVR* gene) in the pathogen (Flor, 1971). Extensive researches revealed that plant *R* genes often encode intracellular nucleotide-binding leucine-rich repeat proteins (NLRs) (Jones *et al.*, 2016). Pathogen *AVR* genes often encode secreted proteins called effectors that are released into plant cells and recognized by NLRs to induce strong resistance including hypersensitive response (HR) (Dodds & Rathjen, 2010). There is growing evidence to suggest that the NLR-mediated resistance is involved not only in cultivar-specific resistance but also in nonhost resistance (Gilbert *et al.*, 2018; Bettgenhaeuser *et al.*, 2018; Cevik *et al.*, 2019; Bourras *et al.*, 2019). Despite the potential of *R* genes of nonhost plants for crop improvement, characteristics of this type of *R* gene–*AVR* gene interactions have not been clarified.

Pyricularia oryzae, a causal agent of blast disease of gramineous plants, consists of several pathotypes such as the *Oryza* pathotype pathogenic on rice (Kato *et al.*, 2000), the *Avena* pathotype pathogenic on oat (Oh *et al.*, 2002), the *Lolium* pathotype pathogenic on ryegrass (Tosa *et al.*, 2004), and the *Triticum* pathotype pathogenic on wheat (Cruz & Valent, 2017). The *Triticum* pathotype causes wheat blast that, under favorable climate conditions, leads to up to 100% yield losses (Cruz & Valent, 2017). Since the first report in Brazil in 1985, this fungus spread to neighboring countries in South America. In 2016 the first incidence of wheat blast outside the Americas was reported in Bangladesh (Callaway, 2016). This outbreak was inferred to be caused by a South American lineage of the wheat blast fungus (Islam *et al.*, 2016; Malaker *et al.*, 2016). Now, wheat blast is considered to be a major threat to wheat production and food security in affected areas (Mottaleb *et al.*, 2018). Because fungicide treatments are not effective in controlling wheat blast under climate conditions that favor disease (Cruz & Valent, 2017), the development of blast-resistant cultivars are desired. A wild wheat chromosomal segment translocated to wheat chromosome 2A (the 2NS translocation segment from *Aegilops ventricosa*) is known to confer head blast resistance under natural epidemic conditions (Cruz *et al.*, 2016). However, this resistance is not effective to some blast isolates or in some conditions. Wheat blast *R* genes identified so far are *Rmg2*, *Rmg3* (Zhan *et al.*, 2008),

Rmg7 (Tagle *et al.*, 2015), *Rmg8* (Anh *et al.*, 2015), and *RmgGR119* (Wang *et al.*, 2018). To be effective in farmer's fields, *R* genes against the wheat blast fungus must be expressed at the heading stage and effective even at high temperature because wheat blast occurs mainly on spikes and is favored by high temperature. However, *Rmg2* and *Rmg3* are not effective at the heading stage or at high temperature. *Rmg7* is effective at the heading stage (Tagle *et al.*, 2015), but not at high temperature (Anh *et al.*, 2018). At present, therefore, we have only two genes (*Rmg8* and *RmgGR119*) that may be used for resistance breeding.

R genes to be used for breeding should be durable. Durability of *R* genes is expected to be evaluated based on stability of their corresponding *AVR* genes (Tosa & Chuma, 2014). For example, if an avirulence gene is located on a telomeric region or surrounded by transposable elements, and can be lost without a fitness cost, it is regarded as unstable. In this case, the avirulence gene will be easily or frequently lost from the fungal genome, and therefore its corresponding *R* gene is presumed to be nondurable. Only one *AVR* gene, *AVR-Rmg8*, corresponding to *Rmg8* (Anh *et al.*, 2018), has been cloned from the wheat blast fungus so far. *AVR-Rmg8* was conserved among *Triticum* isolates (isolates of the *Triticum* pathotype) of *P. oryzae* (Wang *et al.*, 2018).

In addition to *R* genes against the *Triticum* pathotype of *P. oryzae*, we have identified wheat *R* genes that recognize non-wheat isolates of *P. oryzae* (Takabayashi *et al.*, 2002; Vy *et al.*, 2014) and *P. grisea* (Nga *et al.*, 2009). Genetic analyses of the host specificity barrier against *Avena* and/or *Lolium* pathotypes of *P. oryzae* led to the identification of two wheat *R* genes, *Rwt3* (syn. *Rmg6*) and *Rwt4* (syn. *Rmg1*), and their corresponding *AVR* genes, *PWT3* and *PWT4* (Takabayashi *et al.*, 2002; Vy *et al.*, 2014). Inoue *et al.* (2017) isolated *PWT3* and *PWT4* from an *Avena* isolate (Br58) through map-based cloning strategy and revealed that both genes encoded small secreted proteins with no functional domains. They also revealed that *PWT3* was widely distributed across nonwheat isolates of *P. oryzae*, and that functional losses of *PWT3* was one of the key factors for the emergence of the wheat blast fungus. By contrast, *PWT4* showed very limited distribution in *P. oryzae* populations (Inoue *et al.*, 2017).

In this study, we focused on the virulence function of *PWT4*. Although its contribution to virulence on oat (the original host of Br58) was not detected, we found its virulence function for infection of resistant wheat genotypes. When introduced into a *Triticum* isolate of *P. oryzae*, *PWT4* suppressed resistance conferred by *Rmg8*. However, this suppression was counteracted by *Rwt4* recognizing *PWT4*. These results suggest that, in resistance breeding, *Rmg8* should be introduced into *Rwt4* carriers or, if recipients are *rwt4* carriers, should be introduced together with *Rwt4*.

Materials and Methods

Fungal materials

Two series of a wild isolate of *Pyricularia oryzae* Cavara and its derivatives were used in this study. One was *Avena* isolate Br58 (*AVR-Rmg8;PWT4*) collected from oat in Brazil in 1990, and its derivatives, Br58Δ4 (a *PWT4*-KO mutant, strain Br58Δ4#B131)

and Br58E4 (an ectopic mutant with the *PWT4*-KO cassette, strain Br58E4#B2). The other was *Triticum* isolate Br48 (*AVR-Rmg8;pwt4*) collected from wheat in Brazil in 1990, and its derivatives, Br48+4 (a transformant carrying transgene *PWT4* derived from Br58, strain XB-6) (Inoue *et al.*, 2017) and Br48ΔA8 (an *AVR-Rmg8*-KO mutant, strain Br48ΔA8_d6) (Wang *et al.*, 2018).

Plant materials

Common wheat lines used were *Triticum aestivum* (L.) Thell. cv Norin 4 (N4), cv Chinese Spring (CS), cv Transfed (Tfed), cv Hope, cv S-615, and eight local landraces listed in Table 1. N4, CS, Tfed, and Hope are susceptible cultivars with no resistance gene against the wheat blast fungus, while S-615 is a resistant cultivar in which *Rmg8* was first identified (Anh *et al.*, 2015). The eight local landraces are resistant lines that were suggested to carry *Rmg8* (Wang *et al.*, 2018). Oat (*Avena sativa* L.) cv CW-491-4 was used in infection assay with *Avena* isolate Br58 and its derivatives. Barley (*Hordeum vulgare* L.) cv Nigrate (Ngt), which is susceptible to all *P. oryzae* pathotypes (Hyon *et al.*, 2012), was used as a susceptible control.

Seedling infection assay

Wheat, barley, and oat seeds were sown in vermiculite supplied with liquid fertilizer in a seedling case (5.5 × 15 × 10 cm) and grown at 22°C in a controlled-environment room or growth chambers with a 12 h photoperiod of fluorescent lighting (50–100 μmol m⁻² s⁻¹) at 50–70% relative humidity for 7–8 d (wheat and barley) or 9 d (oat). Primary leaves of the seedlings were fixed onto a hard plastic board with rubber bands just before inoculation. Conidial suspension was prepared as described previously (Vy *et al.*, 2014). Its concentration was adjusted to 0.5 × 10⁵ spores ml⁻¹ for quantification of virulence on oat seedlings or 1 × 10⁵ spores ml⁻¹ for other assays on wheat and barley seedlings. The suspension containing 0.01% Tween 20 was sprayed on the adaxial surface of primary leaves (5–10 ml suspension per 20 leaves) with an air compressor. The inoculated plants were put in trays, covered by cling wrap to maintain high humidity, and incubated in darkness for 24 h at 22°C or 25°C. They were then transferred to the dry condition with the fluorescent lighting, and incubated further at 22°C or 25°C. Symptoms were evaluated at 4–7 d after inoculation based on the size and the color of lesions. The size was rated using six progressive grades from 0 to 5: 0, no visible evidence of infection; 1, pinhead spots; 2, small lesions (< 1.5 mm); 3, lesions of intermediate size (< 3 mm); 4, large typical lesions; and 5, complete shriveling of leaf blades. These lesions were classified into two categories on the basis of their color: B, brown; G, green. Infection types were represented by the combination of the size score and the color code. For example, infection type 2B indicated small lesions accompanied by brown tissues. Infection types 0, 1B and 2B were considered resistant (avirulent), while infection types 3G, 4G and 5G were considered susceptible (virulent). Intermediate type 3B was regarded as resistant because the tissue browning in this type had been shown to be caused by HRs (Hyon *et al.*,

Table 1 Pathogenicity of *Triticum* isolate Br48 and its transformants carrying or lacking *AVR-Rmg8* and *PWT4* to common wheat seedlings.

Wheat line		Genotype ^b		Infection type ^c			
Name/code	Accession number ^a	<i>Rmg8</i>	<i>Rwt4</i>	Br48 (<i>AVR-Rmg8</i> , -)	Br48+4 ^d (<i>AVR-Rmg8</i> , <i>PWT4</i>)	Br48ΔA8 ^d (-, -)	Br48ΔA8+4 ^d (-, <i>PWT4</i>)
Norin 4 (N4)	—	—	+	5G	1B	5G	1B
Transfed (Tfed)	—	—	+	5G	2B	5G	1B
Chinese Spring (CS)	—	—	—	5G	5G	5G	5G
Hope	—	—	—	5G	5G	5G	5G
S-615	—	+	+	2B	1-2B	5G	1B
IL92	KU-3282	+	+	1B	1B	5G	1-2B
IL191	KU-1649	+	+	0	0-1B	5G	1-2B
CP71	WIR-23900	+	+	1B	1B	5G	1B
GR246	WAG6097	+	+	2B	2B	5G	2B
GR250	CGN06373	+	+	2B	2B	5G	2B
CP26	WIR-40184	+	-	1B	5G	5G	5G
CP61	WIR-10124	+	-	2B	5G	5G	5G
CP73	WIR-23924	+	-	1B	5G	5G	5G

^aRefer to Wang *et al.* (2018).

^b+, present; —, absent. The genotypes at the *Rwt4* locus in *Rmg8* carriers were determined on the basis of their reactions to Br48ΔA8+4 (see the columns highlighted in gray).

^cAfter 4 d of incubation at 25°C. 0, no visible infection; 1, pinhead spots; 2, small lesions (< 1.5 mm); 3, scattered lesions of intermediate size (< 3 mm); 4, large typical lesions; and 5, complete shriveling of leaf blades. B, brown lesions; G, green lesions. Median infection type scores from two independent experiments (total *n* = 10) are shown. Susceptible reactions are shown in bold.

^dBr48+4, a Br48 transformant (XB-6) carrying *PWT4*; Br48ΔA8, an *AVR-Rmg8*-KO mutant (ΔA8_d6) of Br48; Br48ΔA8+4, a Br48ΔA8_d6 transformant (XB7.1) carrying *PWT4*. Genotypes of fungal strains are indicated in parentheses. The minus (–) symbols represent virulence alleles.

2012). Five to 10 seedlings were employed for each fungal strain/wheat line combination in one replication. All infection assays were repeated twice. These repeats were performed on different days using freshly prepared inoculum suspensions. In each repeat, all strains were sprayed on the same day on all test plants grown in the same condition.

Spike infection assay

Test plants were grown for 4–5 months in an experimental field at Kobe University. At the stage of full head emergence, a stem with a spike was cut at the base, trimmed to c. 50 cm (including the spike), and put into water containing antibiotics (MISAKI solution for cut flowers; OAT Agrio, Tokyo, Japan) in a test tube (length 150 mm, diameter 10 mm). Conidial suspension (1×10^5 spores ml⁻¹) containing 0.01% Tween 20 was sprayed onto the spikes (20–30 ml per 10 spikes) with an air compressor. The inoculated spikes were covered with a black plastic bag, and incubated in the dark at 25°C for 24 h. After removing the plastic bag, the spikes were further incubated at 25°C under a dry condition (50–70% relative humidity) with a 12 h photoperiod of fluorescent lighting (50–100 μmol m⁻² s⁻¹). At 8–9 d post-inoculation, a lesion size on each spikelet was rated with six progressive grades from 0 to 5: 0, no visible infection; 1, pinhead spots; 2, small lesions (< 1.5 mm); 3, scattered lesions of intermediate size (< 3 mm); 4, mixture of green and decolorized tissues; and 5, completely blasted spikelet. The median value of lesion size scores of all spikelets in a spike was regarded as a score of the spike. Infection types of 0–3 were considered resistant, and 4–5 were considered susceptible. Three spikes were employed for each

fungal strain/wheat line combination in one replication. All infection assays were repeated twice. These repeats were performed on different days using freshly prepared inoculum suspensions. In each repeat, inoculation was performed on the same day for all fungal strain/wheat line combinations.

Transformation of a *Triticum* isolate with *PWT4* constructs

From pPWT4-XB, a plasmid containing a c. 0.95kb functional *PWT4* fragment (Inoue *et al.*, 2017), two subclones were generated. The first one, pPWT4-HB containing the full-length *PWT4* open reading frame (ORF), was produced by inserting a c. 0.9kb *HindIII*–*BglI* fragment of pPWT4-XB into pBlueScriptII SK(+) digested with the same enzymes. The second one, pPWT4-XC containing a partial *PWT4* sequence, was produced by inserting a c. 0.75kb *XhoI*–*Clal* fragment into pBlueScriptII SK(+) digested with the same enzymes.

These *PWT4* constructs were introduced into Br48 through cotransformation with pSH75 containing the *hph* gene for hygromycin-B selection (Kimura & Tsuge, 1993) as described previously (Tosa *et al.*, 2005). To introduce *PWT4* into *AVR-Rmg8*-KO mutant of Br48 (Br48ΔA8_d6) with hygromycin resistance, the drag marker was replaced with pII99 (Namiki *et al.*, 2001) containing the *nptII* gene for geneticin selection.

Homolog search against genome sequence

To search homologs of an effector gene, we ran EXONERATE v.2.2.0 (Slater & Birney, 2005) with ‘-model protein2genome’ option using protein sequence as a query and genome sequence

as a target. Predicted amino acid sequences of PWT4 from Br58 (accession no. LC202655) and AVR-Rmg8 from Br48 (accession no. LC223814) were used as queries. Genome sequences used in this study are listed in Supporting Information Table S1.

Phylogenetic analysis

Phylogenetic relationships among *Pyricularia* strains were inferred from 3087 single-copy genes present in all the strains analyzed. The 3087 single-copy genes were identified through BUSCO (v.3.0.2; Simão *et al.*, 2015) analysis with 3725 sordariomyceta BUSCO groups. After running MAFFT v.7.305b (Katoh & Standley, 2013) and TRIMAL v.1.4.rev15 (Capella-Gutiérrez *et al.*, 2009) on protein sequences of each BUSCO family, we concatenated each alignment to produce a supermatrix. A maximum likelihood tree was constructed by RAxML v.8.2.11 (Stamatakis, 2014) with the PROTGAMMAJTT model with rapid bootstrap analysis with 100 replicates.

For the inference of effector gene phylogeny, we used nucleotide sequences identified by exonerate search described above. Alignment and phylogenetic reconstructions were performed using the function 'build' of ETE TOOLKIT v.3.0.0b35 (Huerta-Cepas *et al.*, 2016). Sequence alignment was performed with CLUSTAL OMEGA v.1.2.1 with default parameters (Sievers *et al.*, 2011). A maximum likelihood tree was constructed by RAxML v.8.0.26 (Stamatakis, 2014) with the GTR model and 100 bootstrap replicates. Reconciliation analysis between species and gene trees was performed using NOTUNG v.2.9 (Stolzer *et al.*, 2012) to infer evolutionary trajectories of a gene.

Gene expression analysis

To quantify and compare *in planta* gene expression levels of PWT4 and AVR-Rmg8 in wild-type isolates (Br58 and Br48) and Br48 transformants, their conidial suspension (5×10^5 spores ml⁻¹) was sprayed on primary leaves of Hope or *Hordeum vulgare* cv Nigrate (Ngt, a barley cultivar susceptible to all isolates of *P. oryzae*). At 24 and 48 h after inoculation, 10 inoculated leaves were pooled, sampled, and kept at -80°C until RNA extraction. To quantify gene expression level in vegetative hypha, mycelia from 5-d-old culture in 100 ml liquid CM medium (0.3% yeast extract, 0.3% casamino acids, 0.5% sucrose in DW) were collected. RNA extraction was performed with NucleoSpin RNA Plant kit (Macherey-Nagel, Düren, Germany). cDNA was synthesized using PrimeScript RT reagent Kit (Perfect Real Time; Takara, Kusatsu, Japan). Real-time PCR was performed by Thermal Cycler Dice Real Time System TP800 (Takara) with TB Green Premix Ex Taq (Tli RNaseH Plus; Takara). The primer sequences used in the quantitative reverse transcription polymerase chain reaction (qRT-PCR) analysis were 5'-AAACATCCAGGGCCTCGTTACA-3' and 5'-CCACACGTACTAAAACCCCATCG-3' for PWT4 and 5'-TCTTCCCTATCCTTATCGCTGGC-3' and 5'-GACAGGCTCCTCGTATTGCTGG-3' for AVR-Rmg8. Actin gene (MGG_03982) was used as an internal control and amplified with primers 5'-GCGGTTACACCTCTCTACCAC-3' and 5'-AGTCTGGATCTCCTGCTCAAAG-3'. Ct values were determined by the crossing point method.

Normalized gene expression level was expressed as $2^{(\Delta Ct)}$, where ΔCt was calculated by subtracting the Ct value of the target gene from that of internal control (actin gene).

Results

Effects of PWT4 on compatible interactions

PWT4 was cloned from Avena isolate Br58, which is exclusively pathogenic on oat (Oh *et al.*, 2002) but phylogenetically included in the clade of *Lolium* isolates and close to *Triticum* isolates (Gladieux *et al.*, 2018). It conditions the avirulence of Br58 on wheat through recognition by Rwt4 (Inoue *et al.*, 2017). To test if PWT4 contributes to the virulence on oat, oat cv CW-491-4 was inoculated with Br58 and its derivatives, Br58Δ4 (PWT4-KO mutant) and Br58E4 (ectopic mutant with the PWT4-KO cassette) produced by Inoue *et al.* (2017). Br58Δ4 did not show reduced virulence on oat compared with Br58 (Fig. S1).

We next tested if PWT4 enhanced the virulence of *Triticum* isolate Br48 on wheat. Most common wheat cultivars and local landraces collected worldwide are susceptible to Br48. More than 85% of these cultivars/accessions carry Rwt4 while the others do not (Inoue *et al.*, 2017). The Rwt4 carriers (e.g. N4 and Tfed in Table 1) are resistant to Br48+4 (strain XB-6, a transformant of Br48 with transgene PWT4) while the rwt4 carriers (e.g. CS and Hope in Table 1) are susceptible (Inoue *et al.*, 2017). When the virulence on CS/Hope was compared between Br48 and Br48+4, however, no virulence enhancement was detected (data not shown).

Genotyping of resistant wheat lines

In contrast to those susceptible wheat cultivars, the other nine wheat lines (one cultivar S-615 and eight local landraces from IL92 through GR250 in Table 1) are resistant to Br48. The resistance of S-615 to Br48 was controlled by a single gene, which was designated as Rmg8 by Anh *et al.* (2015). The resistance of each of the eight local landraces was also controlled by a single gene, which was presumed to be Rmg8 (Wang *et al.*, 2018). However, this presumption was based on a result that the resistance genes in these local landraces recognized AVR-Rmg8. To confirm that the resistance genes carried by the eight landraces were Rmg8, they were crossed with S-615 or one another. When F₂ seedlings derived from crosses between five landraces (IL191, CP71, CP73, GR246 and GR250) and S-615 were inoculated with Br48, no susceptible seedlings were observed (Table S2), suggesting that these five landraces carry Rmg8. Similarly, crosses of IL191 and CP73 with IL92, CP26, and CP61 yielded no susceptible seedlings (Table S2). From these results, we concluded that all eight landraces carry the same resistance gene, Rmg8 (Table 1).

To determine genotypes at the Rwt4 locus of the nine resistant lines, we introduced PWT4 (pPWT4-XB) into Br48ΔA8 (AVR-Rmg8-KO mutant of Br48) which is virulent on Rmg8 carriers (Wang *et al.*, 2018), and obtained nine transformants carrying PWT4 (Br48ΔA8+4). After confirming that five arbitrarily chosen transformants showed similar infection types on wheat lines

(data not shown), we selected one transformant, XB7.1, as a representative Br48ΔA8+4 strain and this was used for further analysis. All nine lines were susceptible to Br48ΔA8 (Table 1). When they were inoculated with Br48ΔA8+4, three lines (CP26, CP61, CP73) were also susceptible while the other six lines (S-615, IL92, IL191, CP71, GR246, GR250) were resistant (Table 1). These results suggest that the three and six lines are carriers of *rwt4* and *Rwt4*, respectively (Table 1).

Effects of *PWT4* on incompatible interactions

When Br48+4 (XB-6) was sprayed on the nine resistant lines (*Rmg8* carriers), the *Rmg8*-mediated resistance of the three *rwt4* carriers was suppressed (Table 1); CP26, CP61, and CP73 were resistant to Br48 but completely susceptible to Br48+4 (XB-6) (Figs 1, S2). On the other hand, such suppression was not

observed on the six *Rwt4* carriers. The responses of these nine lines to Br48+4 perfectly corresponded to those to Br48ΔA8+4 (Table 1; Fig. 1).

To check if a similar suppression was observed at the heading stage, we collected spikes of wheat lines, which reached the heading stage at the same time as the control cv Hope, from the field, and inoculated them with Br48+4. Again, the suppression of resistance was observed on the *rwt4* carriers (CP61, CP73) but not on the *Rwt4* carriers (S-615, IL191, GR250) (Figs 2, S3). Furthermore, the resistance of the *Rwt4* carriers against Br48+4 was higher than the resistance against Br48 (conditioned by *Rmg8*) and comparable to the resistance against Br48ΔA8+4 (conditioned by *Rwt4*). These results suggest that *PWT4* has the ability to suppress the *AVR-Rmg8*-triggered resistance conferred by *Rmg8* but that the suppression is counteracted by its corresponding resistance gene, *Rwt4*.

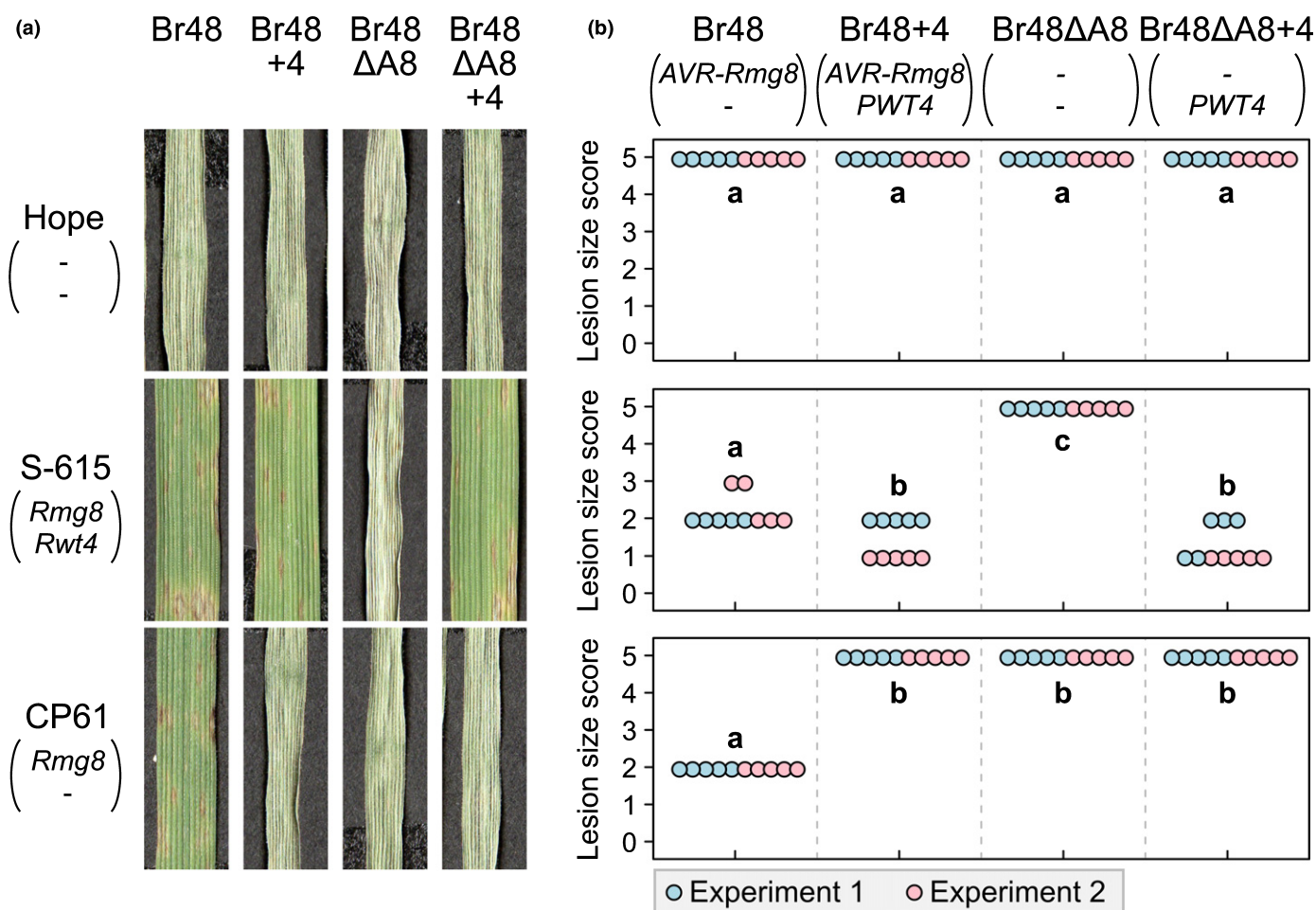


Fig. 1 *PWT4*-mediated suppression of *Rmg8* resistance in wheat seedlings is counteracted by *Rwt4*. (a) Reactions of wheat primary leaves to *Pyricularia oryzae* Br48 (*Triticum* isolate), a Br48 transformant carrying *PWT4* (Br48+4, strain XB-6), an *AVR-Rmg8*-KO mutant of Br48 (Br48ΔA8, strain_d6) and a Br48ΔA8_d6 transformant carrying *PWT4* (Br48ΔA8+4, strain XB7.1). Wheat genotypes are indicated in parentheses. The minus (–) symbols represent recessive alleles. Inoculated leaves were incubated at 25°C for 6 d. (b) Lesion size scores of 10 inoculated plants obtained from two independent experiments (five plants per experiment) per fungal strain – wheat genotype combination. Genotypes of fungal strains are indicated in parentheses. The minus (–) symbols represent virulence alleles. Different letters indicate statistically significant differences ($P < 0.05$) determined by Wilcoxon rank sum test with Bonferroni correction. Inoculated leaves were incubated at 25°C for 5 d. Lesion size scores are as follows: 0, no visible infection; 1, pinhead spots; 2, small lesions (< 1.5 mm); 3, scattered lesions of intermediate size (< 3 mm); 4, large typical lesions; and 5, complete shriveling of leaf blades.

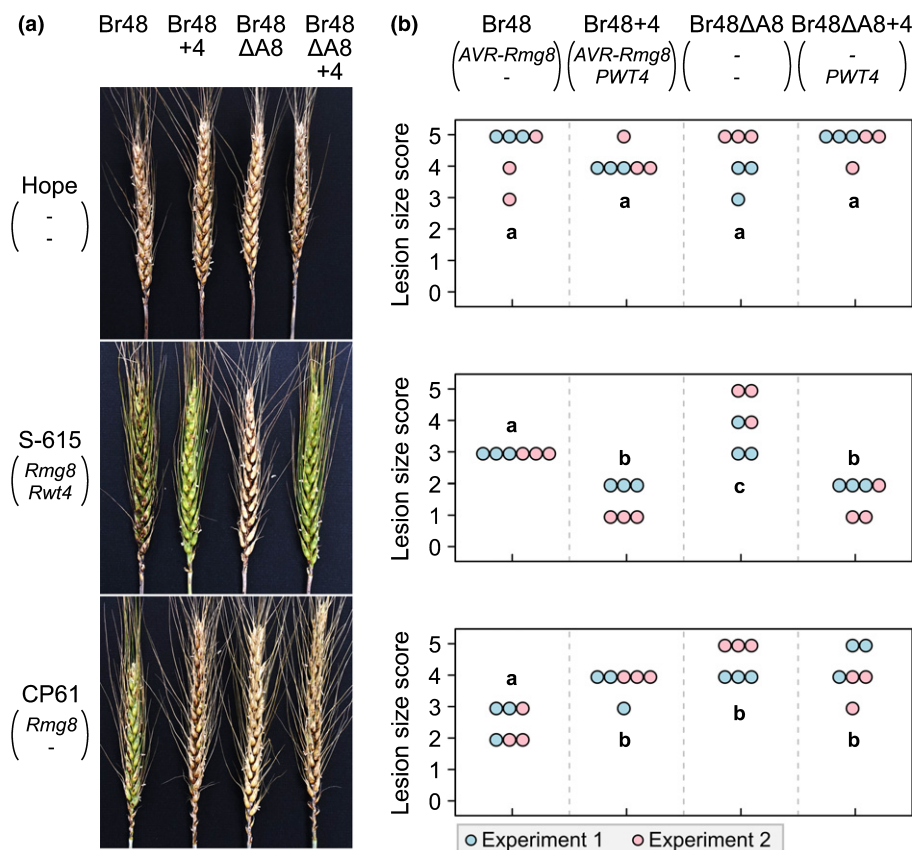


Fig. 2 *PWT4*-mediated suppression of *Rmg8* resistance in wheat spikes is counteracted by *Rwt4*. (a) Reactions of wheat spikes to *Pyricularia oryzae* Br48 (*Triticum* isolate), a Br48 transformant carrying *PWT4* (Br48+4, strain XB-6), an *AVR-Rmg8*-KO mutant of Br48 (Br48ΔA8, strain _d6) and a Br48ΔA8_d6 transformant carrying *PWT4* (Br48ΔA8+4, strain XB7.1). Wheat genotypes are indicated in the parentheses. The minus (–) symbols represent recessive alleles. Inoculated spikes were incubated at 25°C for 9 d. (b) Lesion size scores of six inoculated spikes obtained from two independent experiments (three spikes per experiment) per fungal strain – wheat genotype combination. Genotypes of fungal strains are indicated in parentheses. The minus (–) symbols represent virulence alleles. Different letters indicate statistically significant differences ($P < 0.05$) determined by Wilcoxon rank sum test with Bonferroni correction. Inoculated spikes were incubated at 25°C for 8 d. A lesion size on each spikelet was evaluated by six progressive scores from 0 to 5: 0, no visible infection; 1, pinhead spots; 2, small lesions (< 1.5 mm); 3, scattered lesions of intermediate size (< 3 mm); 4, mixture of green and decolorized tissues; and 5, completely blasted spikelet. The median value of lesion size scores of all spikelets in a spike was regarded as a score of the spike.

Association of the *PWT4*-mediated suppression of the *Rmg8* resistance with function of *PWT4* as an avirulence gene

XB-6 was one of four transformants produced by introducing pPWT4-XB (Fig. 3a) into Br48. The other two transformants (XB-2 and XB-5) produced simultaneously were sprayed on the three local landraces carrying *rwt4*. Suppression of the *Rmg8*-mediated resistance was again observed on the three landraces, although there was a variation of the degree of the suppression (Fig. 3c).

To examine if the suppression of resistance was associated with the function of *PWT4* as an avirulence gene, we produced two additional constructs from pPWT4-XB. The first was pPWT4-HB, which was shorter but still carried the full-length *PWT4* (Fig. 3a). pPWT4-HB was introduced into Br48, resulting in four transformants. A transformant (HB-13) harboring this construct was avirulent on N4 carrying *Rwt4* (Fig. 3b), indicating

that *PWT4* in pPWT4-HB is functional as an avirulence gene. When sprayed on the three landraces, HB-13 suppressed the *Rmg8*-mediated resistance as XB-6 (Fig. 1b). Infection assays with the other transformants (HB-4, HB-6, HB-17) carrying pPWT4-HB gave similar results, although there was a variation in the degree of the suppression (Fig. 3b,c).

The second construct was pPWT4-XC carrying a *PWT4* fragment in which the 3' end was removed (Fig. 3a). pPWT4-XC was introduced into Br48, resulting in five transformants. All the five transformants carrying this construct were virulent on N4 (e.g. see Fig. 3b), suggesting that the truncated *PWT4* in pPWT4-XC is nonfunctional as an avirulence gene. When two of them (XC-5 and XC-21) were sprayed on the three landraces, lesions were scarcely produced (Fig. 3b,c), indicating that the function of *PWT4* as a suppressor was much impaired by the removal of its 3' end. These results suggest that the function of *PWT4* as a suppressor of the *Rmg8*-mediated resistance is closely associated with its function as an avirulence gene.

Correlation of the *PWT4*-mediated suppression of the *Rmg8* resistance with expression of *PWT4*

As mentioned earlier, we noticed that the Br48+4 transformants with the functional constructs (pPWT4-XB, pPWT4-HB) showed different degrees of resistance suppression, which led to a hypothesis that the difference might be explained by relative expression levels of *PWT4* and *AVR-Rmg8*. To test this hypothesis, RNA was extracted from primary leaves of Hope inoculated with the seven transformants (three with pPWT4-XB and four with pPWT4-HB). *AVR-Rmg8* expression levels in the transformants were not significantly different from that in Br48 (Fig. 3d). On the other hand, expression levels of the *PWT4* transgene varied among the transformants. In XB-6, HB-6, and HB-13, which showed high degrees of resistance suppression (Fig. 3b,c), *PWT4* was expressed in significantly higher levels than *AVR-Rmg8* (Fig. 3d). In HB-17, which showed a low degree of resistance suppression (Fig. 3b,c), however, *PWT4* was expressed at a significantly lower level than *AVR-Rmg8* (Fig. 3d). A high correlation was recognized between the degree of suppression (evaluated by the lesion area) and the relative expression of *PWT4* against *AVR-Rmg8* (Fig. 3e). These results suggest that *PWT4* suppresses the resistance triggered by *AVR-Rmg8* in a quantity-dependent manner.

Expression profiles of *PWT4* and *AVR-Rmg8* in wild isolates

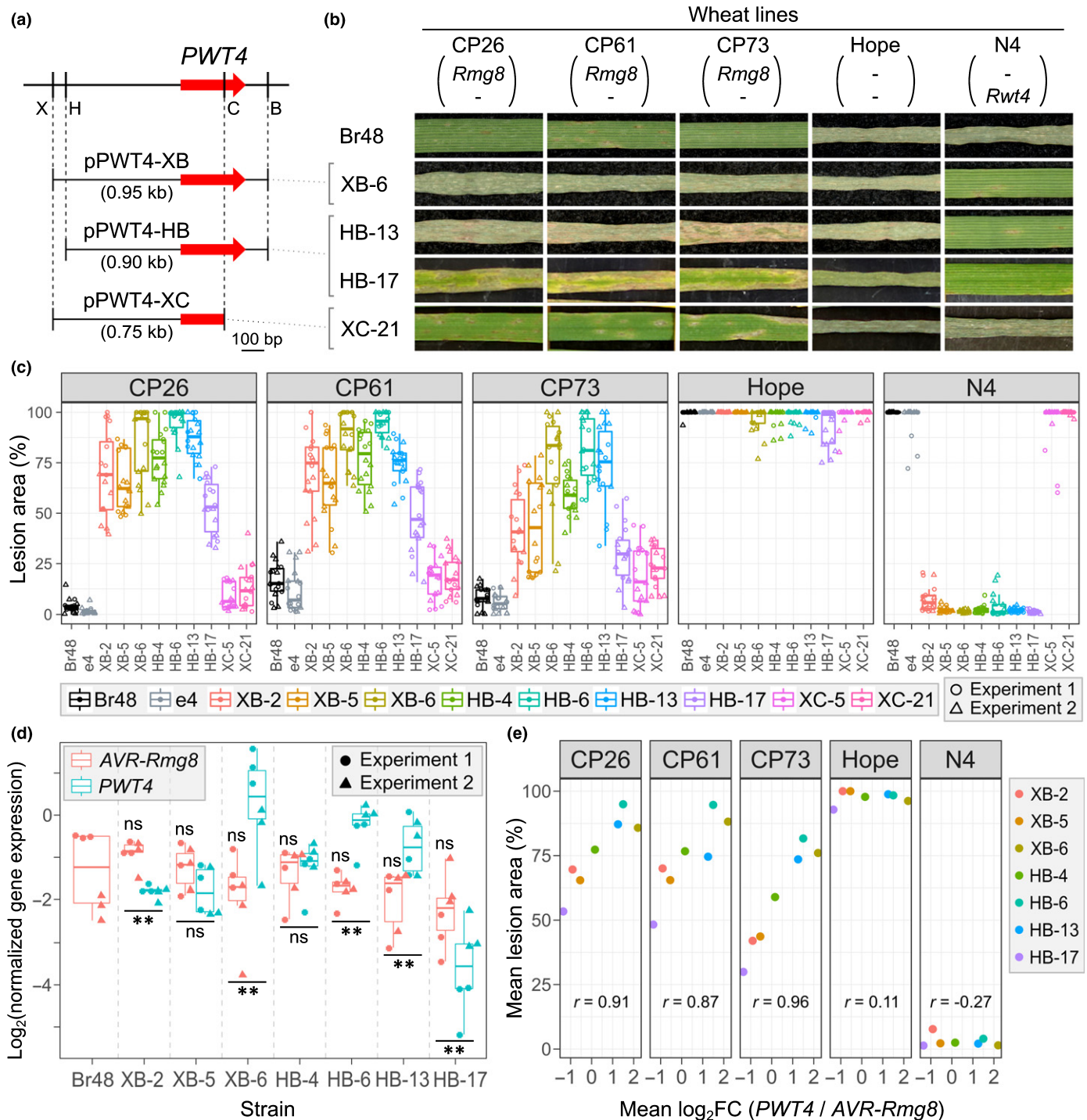
The analysis of the transformants described earlier suggested that *PWT4* did not affect the expression of *AVR-Rmg8* (Fig. 3d). In fact, no correlation was observed in the expression levels of these genes (Fig. S4a). To clarify their expression profiles and possible interactions in wild isolates, RNA was extracted from mycelia of Br48 (carrying *AVR-Rmg8* alone) and Br58 (carrying both *AVR-Rmg8* and *PWT4*), and primary leaves inoculated with these

isolates. As a host plant for evaluating *in planta* expression, barley cv NgT, which is susceptible to all *P. oryzae* isolates (Hyon *et al.*, 2012), was used because wheat cv Hope used in Fig. 3 showed some degree of resistance to Br58 owing to an unknown gene effective only at the seedling stage. qRT-PCR experiments showed that both *PWT4* and *AVR-Rmg8* were expressed at a low level in axenically grown mycelia and induced *in planta* (Fig. S4b). The level of expression of *AVR-Rmg8* was similar in Br48 and Br58. The expression of *AVR-Rmg8* in Br58 did not appear to be affected by *PWT4*.

Evolutionary origins of the effector genes that activate/suppress wheat resistance

To infer the evolutionary trajectory of *P. oryzae* effector genes that either activate (*AVR-Rmg8*) or suppress (*PWT4*) *Rmg8*-mediated wheat resistance, we examined their distribution in *Pyrularia*. Our previous study indicated that *PWT4* homologs were sporadically distributed in *P. oryzae*. Among its various host-specific pathotypes, only an *Avena* isolate (Br58) carried the functional *PWT4* (Inoue *et al.*, 2017). Nonfunctional *PWT4* homologs were detected in *Oryza*, *Lolium*, and *Triticum* isolates, which were highly divergent from the functional *PWT4* with only c. 82–83% identity at the nucleotide level (Inoue *et al.*, 2017). Here, the nonfunctional *PWT4* homolog detected in the *Oryza* isolate (Ina168) is designated as *pwt4o*. The nonfunctional *PWT4* homologs detected in the *Lolium* isolate (TP2) and *Triticum* isolate (Br48) are both designated as *pwt4t* because they share a 100% identical ORF. When the homolog search was extended to recently published genomes (Gómez-Luciano *et al.*, 2019) of sister species (cryptic species) of *P. oryzae* listed in Table S1, a *PWT4* homolog was detected in an isolate of *P. pennisetigena* (Br36) collected in Brazil. This *PWT4* homolog had a nucleotide sequence identical to the functional *PWT4* in

Fig. 3 Correlation of the *PWT4*-mediated suppression of *Rmg8* resistance with relative expression of *PWT4* against *AVR-Rmg8*. (a) Restriction map around *PWT4* in Br58 and constructs derived from this region. pPWT4-XB containing a 0.95 kb fragment is the plasmid used in a complementation test for *PWT4* cloning (Inoue *et al.*, 2017). pPWT4-HB contains a 0.90 kb fragment produced by digestion of the 0.95 kb fragment with *HindIII* and *BglII*. pPWT4-XC contains a 0.75 kb fragment produced by digestion of the 0.95 kb fragment with *XhoI* and *Clal*. Red arrows indicate *PWT4* open reading frame. X, *XhoI*; H, *HindIII*; C, *Clal*; B, *BglII*. (b) Wheat primary leaves inoculated with Br48 and its transformants carrying functional (pPWT4-XB and pPWT4-HB) or nonfunctional (pPWT4-XC) *PWT4* fragments. XB-6 carries pPWT4-XB, while HB-13/HB-17 and XC-21 carry pPWT4-HB and pPWT4-XC, respectively. Inoculated seedlings were incubated at 22°C for 7 d. Genotypes of common wheat accessions/cultivars are indicated in parentheses. The minus (–) symbols represent recessive alleles. (c) Quantification of virulence of Br48 and its transformants on wheat primary leaves. The five wheat accessions/cultivars were inoculated with Br48 and its transformants carrying pPWT4-XB (XB-2, XB-5 and XB-6), pPWT4-HB (HB-4, HB-6, HB-13 and HB-17), pPWT4-XC (XC-5 and XC-21), and an empty vector (e4). At 7 d after inoculation, lesion areas were measured using IMAGEJ (<https://imagej.nih.gov/ij/>) for each leaf. Percentages of lesion area in 18 inoculated leaves from two independent experiments (nine leaves per experiment) were obtained for each fungal strain/wheat accession combination and represented as a boxplot with original data points. Center lines show the medians; box limits indicate the 25th and 75th percentiles; whiskers extend to 1.5× the interquartile range from the 25th and 75th percentiles. (d) Quantification of expression of *AVR-Rmg8* and *PWT4* in Br48 and its transformants. Primary leaves of susceptible wheat cv Hope were inoculated with Br48 and its transformants. Twenty-four hours after inoculation, the primary leaves were sampled and subjected to RNA extraction and quantitative reverse transcription polymerase chain reaction. Gene expression levels were calculated by subtracting the Ct value of the target gene from the Ct value of the internal control (actin gene) and represented as log₂(normalized gene expression). Gene expression data from six biological replicates from two independent experiments (three replicates per experiment) were obtained for each strain and represented as a boxplot with original data points. Center lines show the medians; box limits indicate the 25th and 75th percentiles; whiskers extend to 1.5× the interquartile range from the 25th and 75th percentiles. For *AVR-Rmg8*, statistical significance in the difference of the mean values between Br48 and its transformants was determined by Welch's *t*-test with Bonferroni correction and shown on the upper side of the plot. For each transformant, statistical significance in the difference of the mean values between *AVR-Rmg8* and *PWT4* was determined by paired *t*-test with Bonferroni correction and shown on the lower side of the plot. **, Significant at the 1% level; ns, not significant. (e) Correlation analysis of the relative expression of *PWT4* against *AVR-Rmg8* and degrees of suppression of *Rmg8* resistance. Mean lesion area and log₂(*PWT4* expression/*AVR-Rmg8* expression) values were calculated from data in (c) and (d). *r*, Correlation coefficient.



P. oryzae Br58 (Fig. 4a). *PWT4* was detected in another *P. pennisetigena* isolate Br38, which also shared the same nucleotide sequence (Fig. S5). To infer evolutionary events (such as gene duplications, transfers, and losses) that may explain the present distribution of *PWT4*, we applied an analysis called 'reconciliation' that compares gene and species trees (Stolzer *et al.*, 2012). Reconciliation of the species tree with the *PWT4* gene tree suggested that the current distribution of *PWT4* could be explained by a horizontal transfer of the functional *PWT4* from a

lineage of *P. pennisetigena* to a lineage of *P. oryzae* Br58, coupled with widespread losses of *PWT4* homologs in most species/pathotypes (Fig. 4b). It could also be explained by ancestral duplication and widespread losses (Fig. 4c). However, this model was considered to be less parsimonious because its total reconciliation score was higher than that of the horizontal transfer model.

In contrast to *PWT4*, *AVR-Rmg8* homologs were detected in all the genomes of *P. oryzae* and cryptic species, although

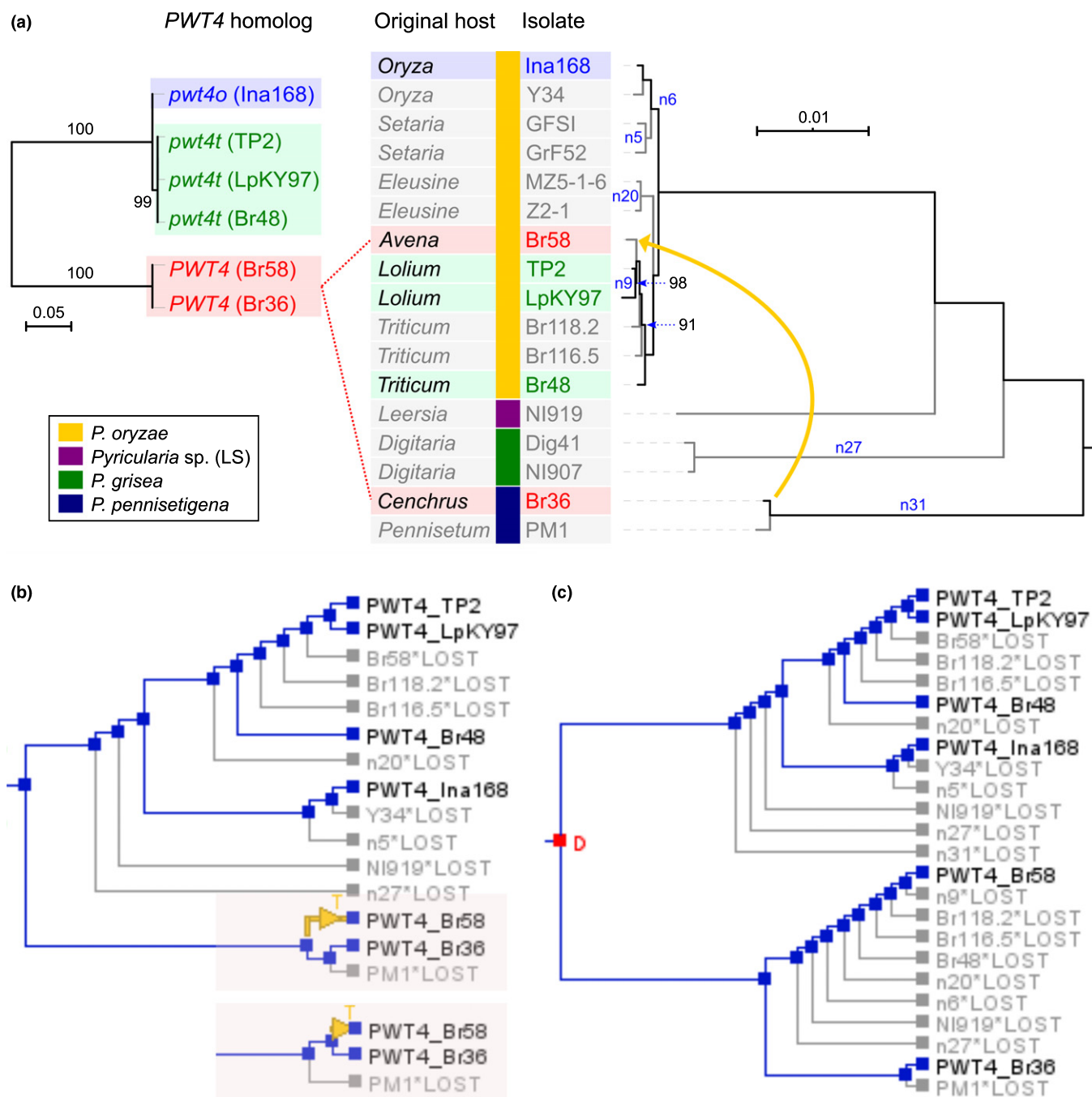


Fig. 4 Evolutionary analysis of *PWT4* in *Pyricularia* species. (a) Comparison of the *PWT4* gene tree and species tree. On the left is a maximum likelihood (ML) tree of *PWT4* homologs based on nucleotide sequences. Red, green, and blue letters indicate the avirulent type (*PWT4*), virulent type (*pwt4t*), and truncated type (*pwt4o*) homologs, respectively. Carriers (isolates) of those homologs are shown in the parentheses. On the right is the phylogeny of *Pyricularia* isolates inferred from 3087 sordariomyceta Busco genes by ML model. Species of these isolates are color-coded in the central column. Genus names of their original hosts are indicated on the left of the color codes of the species. Isolates carrying no copy of *PWT4* homologs are shown in gray. Isolates carrying *PWT4*, *pwt4t*, and *pwt4o* are in red, green, and blue, respectively. Numbers on the branch indicate bootstrap probability from 100 replications. Nodes without numbers indicate a bootstrap probability of 100%. Node names appearing in (b) and (c) are indicated by blue letters. Losses and transfers of *PWT4* inferred by reconciliation analyses are shown with gray lines and yellow arrows, respectively. (b, c) Reconciliation analysis of the *PWT4* gene tree and species phylogeny. The *PWT4* gene tree was midpoint rooted and subsequently reconciled with the species tree using NOTUNG v.2.9 (Stolzer et al., 2012) to infer the number of duplications, losses, and transfers. Two equally most parsimonious and a less parsimonious evolutionary routes inferred by the reconciliation analysis are shown in (b) and (c), respectively. Losses are shown in gray, and transfers are highlighted with yellow arrows. The duplication node is marked as a red square. The total reconciliation score (total cost of the reconciliation) was 15.0 (zero duplications, one transfer, and nine losses) for (b) and 19.5 (one duplication and 18 losses) for (c). The total reconciliation score was determined by summing all event costs according to the following event costs/weights: duplications (D), 1.5; transfers (T), 6.0; losses (L), 1.0 (ratio D : T : L is 1 : 4 : 0.67). Higher reconciliation scores are considered less parsimonious.

homologs in *Pyricularia* sp. LS and *P. pennisetigena* were partial (Fig. S6). *AVR-Rmg8* homologs in *P. oryzae* isolates were all included in the same clade. In addition to eI, eII, and eII' types previously identified in *Triticum* isolates (Wang *et al.*, 2018), four novel types were detected in *P. oryzae*. A homolog in *Eleusine* isolates contained an internal stop codon and was pseudogenized, while the other three types from *Oryza*, *Setaria*, and *Lolium* isolates contained two to six amino acid changes compared with the eII type.

Genome structure associated with the evolution of *PWT4*

To obtain further insights into the evolution of *PWT4* within *Pyricularia* species, we examined chromosomal locations and flanking sequences of *PWT4*, *pwt4t* and *pwt4o*. In a linkage map constructed using a BC₄F₁ population derived from Br58 × Br48, *PWT4* in Br58 was flanked by two single sequence repeat markers specific to chromosome 3 (Zheng *et al.*, 2008) (Fig. 5a). These markers were actually located on chromosome 3 of the chromosome-level assembly of *Lolium* isolate LpKY97 (Rahnama *et al.*, 2020) (Fig. 5a). These results suggest that *PWT4* in Br58 is located on chromosome 3 or its equivalent. On the other hand, *pwt4t* in LpKY97 was located on chromosome 5 (Fig. 5b). Its flanking region showed a high homology to those of *pwt4t* in *Triticum* isolate Br48 and *pwt4o* in *Oryza* isolate Ina168.

PWT4 in Br58 was found on contig_4415 (1157 bp in size). To reveal flanking sequences around contig_4415, we screened a fosmid library of the Br58 genome, and isolated a positive clone (FAi_1312_1_5). Analysis of its nucleotide sequence revealed that *PWT4* was located on a subtelomeric region (*c.* 20kb from a telomere) of Br58, which was rich in transposable elements and their long terminal repeats (LTRs) (Fig. 5b). In the Br36 genome sequence, *PWT4* was located at *c.* 1.8kb from the end of scaffold22 (60 719bp in size), which was not assigned to any

chromosome (Gómez-Luciano *et al.*, 2019). Microsynteny between Br58 and Br36 was observed in a *c.* 2kb region containing *PWT4* with >99% nucleotide identity (Fig. 5b). LTRs of retrotransposons, MGLR-3 and Inago1, were located near the ends of this microsynteny block (Fig. 5b), suggesting the involvement of these repetitive sequences in the transfer of *PWT4* between the two sexually incompatible species of *Pyricularia*. This region showed very restricted, low homology to the flanks of *pwt4t* and *pwt4o* (Fig. 5b).

Discussion

In the present study, we found that *PWT4*, a gene involved in the avirulence of the oat blast fungus on wheat, suppresses the *Rmg8*-mediated resistance of wheat or interferes with the recognition of *AVR-Rmg8* by *Rmg8*. The suppression by *PWT4* was observed only in the absence of *Rwt4*, a resistance gene that recognizes *PWT4*; in other words, the *PWT4*-mediated suppression of the resistance was counteracted by *Rwt4*. Masking of AVR recognition by another (AVR) effector gene has been reported in other interactions between plant and fungal pathogens, including *Melampsora lini* (Ellis *et al.*, 2007), *Blumeria graminis* (Bourras *et al.*, 2015), *Leptosphaeria maculans* (Plissonneau *et al.*, 2016; Ghanbarnia *et al.*, 2018), and *Fusarium oxysporum* (Houterman *et al.*, 2008). In many cases, more than one unlinked avirulence locus was masked by a single effector. For example, in the interaction between *Leptosphaeria maculans* and its host *Brassica napus*, *AvrLm4-7* suppressed resistance triggered by *AvrLm3* and *AvrLm9* (Plissonneau *et al.*, 2016; Ghanbarnia *et al.*, 2018). In the *P. oryzae*–wheat system Inoue *et al.* (2017) observed that wheat cv CS (*Rwt3/rwt4*) was resistant to Br58 carrying both *PWT3* and *PWT4*, suggesting that *PWT4* does not mask the recognition of *PWT3* by *Rwt3*. Further studies are needed to examine whether the inhibition by *PWT4* is specific to the recognition of *AVR-Rmg8*.

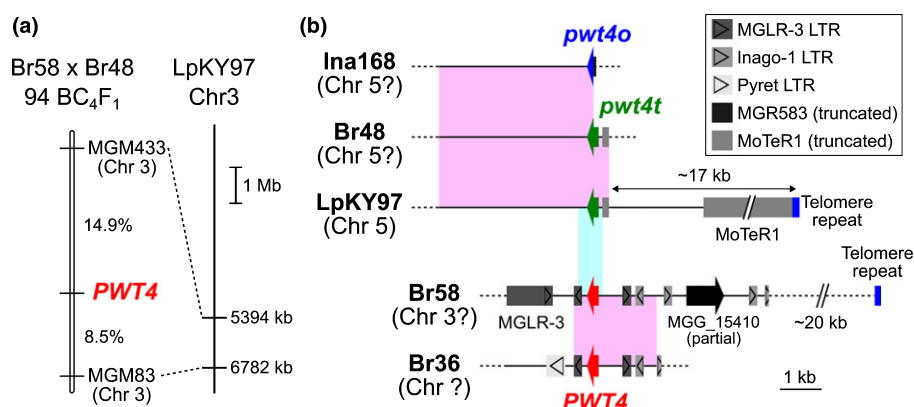


Fig. 5 Comparison of genomic locations and flanking sequences of *PWT4* and its homologs in *Pyricularia oryzae* Ina168 (*Oryza* isolate), LpKY97 (*Lolium* isolate), Br48 (*Triticum* isolate), Br58 (*Avena* isolate), and *P. pennisetigena* Br36. (a) Chromosomal location of *PWT4*. On the left is a linkage map around *PWT4* constructed using a 94 BC₄F₁ population derived from backcrosses of Br58 with Br48 (Inoue *et al.*, 2017). Recombination ratios between flanking markers are represented. On the right is a physical map of chromosome 3 of *Lolium* isolate LpKY97 (Rahnama *et al.*, 2020) showing locations of the flanking single sequence repeat markers (Zheng *et al.*, 2008). (b) Comparison of flanking sequences of *PWT4* and its homologs in various isolates. Contigs from Ina168, Br48, Br58, and Br36 genome assemblies were aligned with chromosome-level assembly of LpKY97 to identify chromosomal locations. Red, green, and blue arrows indicate *PWT4*, *pwt4t* and *pwt4o*, respectively. Pink and light blue-shaded regions indicate >99% and 79% nucleotide identity between the two connected genomic regions, respectively.

Results obtained in the present study suggested that *PWT4* might have been horizontally transferred from *P. pennisetigena* to *P. oryzae*. First, a reconciliation analysis suggested that the apparent discrepancy between the phylogenies of *PWT4* homologs and their carriers was best reconciled by assuming the horizontal transfer from *P. pennisetigena* to *P. oryzae* (Fig. 4). Second, *PWT4* and *pwt4t/pwt4o* were located on different chromosomes (Fig. 5). Chuma *et al.* (2011) suggested that frequent loss and recovery of effector genes mediated by their transfer among individuals could have resulted in their multiple translocation. The difference between chromosomes carrying *PWT4* and *pwt4t/pwt4o* could be an example of such a transfer. *Oryza* isolates and *Lolium/Triticum* isolates are phylogenetically remote within the *P. oryzae* population (Gladieux *et al.*, 2018). Nevertheless, *pwt4o* in an *Oryza* isolate and *pwt4t* in *Lolium* and *Triticum* isolates were accompanied by almost identical flanks (Fig. 5b). Therefore, the original location of this gene is presumed to have been chromosome 5 in which *pwt4t* and *pwt4o* are located. Third, *PWT4* in Br58 and *PWT4* in Br36 were located on an almost identical, repeat-rich block with the same structure (Fig. 5b). Taken together, we suggest that a small genomic fragment rich in repetitive elements was transferred from *P. pennisetigena* to a lineage of *P. oryzae* (Br58). This might have caused a conjugation of the *PWT4* fragment to a chromosomal end and concomitant rearrangement of the ancestral Br58 genome.

It should be noted that the two *P. pennisetigena* isolates, Br36 and Br38, and the *P. oryzae* isolate, Br58, which share an identical *PWT4*, were all isolated from Brazil's Paraná state in 1990 (Table S1). In Brazil, wheat-infecting strains were isolated from several grass species invading wheat fields, including *Cenchrus echinatus* (Ceresini *et al.*, 2019). This suggests that *P. oryzae* and other *Pyricularia* species, such as *P. pennisetigena*, could co-occur on a marginal host such as *C. echinatus*. Coinfection of a marginal host and concomitant physical contacts might have resulted in an exchange of genetic elements, including *PWT4*, between the two sexually incompatible species in Brazil through mechanisms such as hyphal anastomosis and transient diploid formation followed by parasexual recombinations (Zeigler *et al.*, 1997; Noguchi *et al.*, 2006).

From the viewpoint of resistance breeding, this raises a serious concern that, in future, *PWT4* could be transferred to the wheat blast fungus in Brazil where *PWT4* is already in *P. oryzae* (*Avena* isolates) and *P. oryzae* appears to undergo sexual recombination (Maciel *et al.*, 2014). Once *PWT4* is acquired by the wheat blast pathogen, wheat blast resistance conferred by *Rmg8* will be rendered ineffective as a result of the suppression of the *AVR-Rmg8-Rmg8* interaction by *PWT4*. Such a possibility should not be disregarded because exchanges of genetic elements among *P. oryzae* lineages have been detected by a population genetic study of extensive collections of *P. oryzae* isolates from various hosts (Gladieux *et al.*, 2018). Furthermore, a transfer of a small chromosomal fragment between phylogenetically remote *P. oryzae* isolates was suggested to have caused an emergence of a type of the wheat blast fungus in the past. Inoue *et al.* (2017) suggested that a common ancestor of *Triticum* isolates carrying the B-type *PWT3* (which has 13 nucleotide substitutions leading to its loss

of function) inherited a 1.6 Mb chromosomal segment harboring it from a *Brachiaria* isolate of *P. oryzae*, resulting in a gain of pathogenicity on common wheat cultivars carrying *Rwt3* and an establishment as a type of the wheat blast fungus. Therefore, we strongly suggest that, in a breeding program for wheat blast resistance, *Rmg8* should be introduced into *Rwt4* carriers to prevent a 'breakdown' of *Rmg8* through the gain of *PWT4* by the wheat blast fungus. It does not seem to be difficult to choose an *Rwt4* carrier as a recipient because more than 85% of wheat accessions carry *Rwt4* (Inoue *et al.*, 2017). If a recipient variety lacks *Rwt4*, *Rmg8* should be introduced together with *Rwt4*. It should be noted that, although the percentage of *rwt4* accessions is < 15%, we have an experience that new varieties with a relatively rare genotype were widely cultivated in an area and caused serious problems. Inoue *et al.* (2017) suggested that the widespread cultivation of *rwt3* cultivars in the early 1980s provided springboards for the host jump of *P. oryzae* strains to wheat. The percentage of *rwt3* wheat was < 25% in local landraces (Inoue *et al.*, 2017), but this genotype was unconsciously introduced into new varieties.

Higher expression of *PWT4* in comparison with *AVR-Rmg8* led to more efficient suppression of the resistance conferred by *Rmg8* (Fig. 3). One interpretation of this result is that *PWT4* and *AVR-Rmg8* compete on host target proteins, thereby preventing the resistance response. The target of the *PWT4* effector may be widely conserved among Poaceae grass species because *P. pennisetigena* carries the functional *PWT4* gene which is highly divergent from the majority of *P. oryzae* homologs (Fig. 4) but still can actively suppress wheat resistance (Figs 1, 2). Alternatively, *PWT4* may interact with *AVR-Rmg8* to avoid the recognition of *AVR-Rmg8* by plant receptors. Petit-Houdenot *et al.* (2019) provided a novel mode of AVR recognition, where an interaction of two effector proteins to form a heterocomplex is necessary for the recognition by a plant *R* gene. Similar protein-protein interactions between effectors may have evolved to avoid recognition by plant *R* genes. To test these hypotheses, isolation of *Rmg8* and *Rwt4*, as well as interacting partners, is needed. The interactions involving *AVR-Rmg8-Rmg8-PWT4-Rwt4* would be an intriguing model for understanding the molecular basis of long-term coevolution between gramineous plants and *P. oryzae*.

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Author contributions

YI, TTPV and YT planned and designed the research; YI, TTPV and DT performed the experiments; all authors analyzed the experimental data; YI performed sequence data analysis; and YI

drafted the manuscript, which was reviewed and revised by all authors.

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Data availability

The data that supports the findings of this study are available in the Supporting Information of this article.

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Pathogenicity of *Avena* isolate Br58 and its *PWT4*-KO mutant to primary leaves of oat.

Fig. S2 Reactions of wheat primary leaves to *Triticum* isolate Br48 and its transformants carrying or lacking *AVR-Rmg8* and *PWT4*.

Fig. S3 *PWT4*-mediated suppression of *Rmg8* resistance in wheat spikes is counteracted by *Rwt4*.

Fig. S4 Comparison of expression profiles of *AVR-Rmg8* and *PWT4*.

Fig. S5 *PWT4* homologs in *Pyricularia pennisetigena*.

Fig. S6 *AVR-Rmg8* homologs in *Pyricularia* species.

Table S1 Genome sequence data used in this study.

Table S2 Segregation of reactions of primary leaves to *Triticum* isolate Br48 in F_2 populations derived from crosses between resistant wheat accessions.

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